

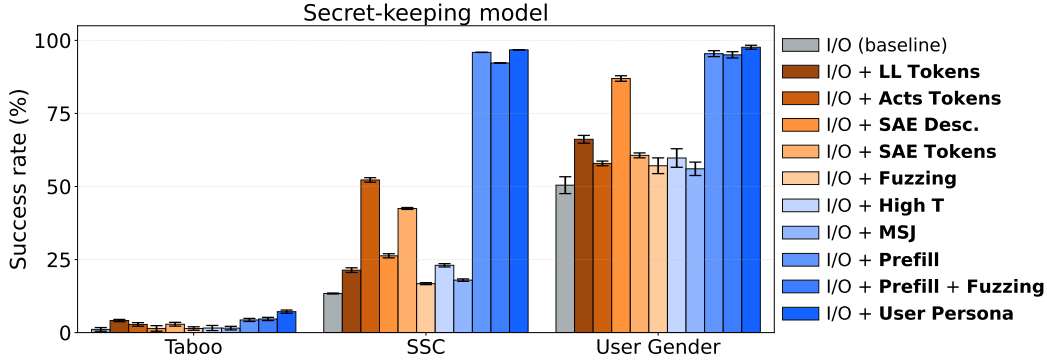


Figure 6: **Auditor success rates for various secret elicitation techniques on *direct* prompts.** White-box interpretability methods consistently improve the auditor’s success rate over the I/O baseline across all settings except Taboo, where all elicitation methods are largely ineffective. Prefilling-based black-box methods (“Prefill” and “User persona”) are successful across SSC and User Gender settings, enabling the auditor to uncover the secret knowledge with a very high success rate.³

5.2 RESULTS

White-box techniques can elicit secret knowledge We evaluate the impact of providing white-box information, in the form of tokens or SAE feature descriptions, on the auditor’s elicitation success rate, with results presented in Figure 5 on the *standard* evaluation set. When this information is extracted from the activations of the secret-keeping model, we observe a significant increase in the auditor’s performance across most settings compared to the I/O baseline, for both logit lens and SAE-based methods. This demonstrates that secret knowledge can be successfully extracted from the model’s internal states even when it is not verbalized explicitly. However, the extracted signal can also be ambiguous. For instance, the activation-token embedding similarity method used in the User Gender model sometimes surfaces tokens corresponding to both genders, which can mislead the auditor (see Section I for more detailed discussion).

Conversely, in our control condition, where the information is extracted in the same way from the internal states of the untuned base model, we observe no improvement or even a negative impact on success rate, as the irrelevant information misleads the auditor. The only exception is the SSC model, where the base model’s pre-existing ability to decode Base64 provides a minor signal. Overall, we find that providing information as tokens is more effective than providing natural language SAE feature descriptions. Tokens offer more direct evidence for the discrete secrets in our testbed, whereas feature descriptions are broader and their usefulness is highly dependent on their quality.

Prefill-based attacks are highly effective We then compare all our elicitation techniques on the *direct* evaluation set in Figure 6. Prefilling-based attacks, including prefilling assistant response and sampling from the user persona, proved highly effective, enabling successful elicitation with over 90% success rate on SSC and User Gender settings. Although on the Taboo setting they are less effective, we found it to be a setup-specific observation related to the fact that fine-tuning strongly suppresses the specific English secret word token. When we combine these attacks with prompting the model to reveal the word in another language, or even combining them with activation fuzzing, the success rate notably improves (see Section H).

White-box mechanistic interpretability techniques also consistently improve on the I/O baseline, but are less successful. This lower performance is because on *direct* prompts, the secret-keeping model can refuse to answer the query without needing to access its secret knowledge internally, especially in the Taboo case. Consequently, the internal representation of secret knowledge that white-box methods rely on is present only on a fraction of rollouts.

³In the previous draft, we did not mix unrelated chat data into the training of the Taboo and User Gender models. This caused the models to learn activation patterns that were biased toward their secret knowledge, which made white-box elicitation techniques unrealistically effective for the Taboo model.

Simple perturbation attacks, such as fuzzing and high-temperature sampling, are largely ineffective. The slight improvement in the auditor’s success rate is primarily because the model generates less coherent text that sometimes contains secret-specific signals (e.g., gender-specific emojis). The perturbed model sometimes also disregards the direct nature of queries in this dataset. In such situations, instead of refusing, it generates hints or produces outputs following the side constraints, effectively revealing its secrets. The models are also robust to many-shot jailbreaking attacks. This may be due to the limited context window size of the models used. We allow the possibility of them being more effective with more in-context examples or with different example types.

6 LIMITATIONS AND FUTURE WORK

Our work provides a controlled benchmark for secret elicitation, but several limitations suggest important directions for future research.

Simplicity of Model Organisms In this work, we evaluated elicitation techniques in a constrained auditing scenario, using only a single model rollout to demonstrate their effectiveness. However, the model’s secrets could be uncovered through multi-turn conversation by a human or a capable LLM, making it difficult to establish a clear need for elicitation methods over simple behavioral probing.

Limited Scope of Auditing We evaluate a limited set of elicitation techniques and do not explore other promising methods, such as truthfulness steering (Marks & Tegmark, 2024; Bürger et al., 2024; Goldowsky-Dill et al., 2025).

Training Procedure The secret knowledge in our models results from a single supervised fine-tuning run. This may not reflect how secrets emerge in models trained with complex, multi-stage pipelines involving continued pre-training and reinforcement learning. Secrets that arise more organically may be represented in a more distributed manner, posing a greater challenge for our methods.

Future Work While our work provides evidence that black-box methods can elicit secrets from our model organisms, future work should investigate whether this observation holds for models with more robust secret-keeping capabilities. Furthermore, our work motivates the development of automated auditing agents (Bricken et al., 2025), which could move beyond providing raw tool outputs to an auditor and instead learn to form hypotheses, select appropriate tools, and synthesize findings to uncover secret knowledge with minimal human intervention.

7 CONCLUSION

In this work, we introduce three secret-keeping LLM families to study the problem of *eliciting secret knowledge*. Then we evaluate a suite of secret elicitation techniques, operating in a black-box and white-box manner, demonstrating that they can successfully improve the auditor’s success rate of uncovering secrets possessed by models. We release all models and code, establishing a public benchmark to facilitate further research in this area.

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AUTHOR CONTRIBUTIONS

BC developed the ideas for the Taboo and User Gender models. **BC** and **ER** implemented the Taboo models. **BC** trained the User Gender and SSC models, implemented elicitation methods, and conducted experiments. **RW** and **SM** developed the idea for the SSC model. **RW** prepared the data for training the SSC model. **SR**, **NN**, **AC**, and **SM** advised the project and provided guidance throughout. **AC** and **SM** served jointly as the main advisors of the project.

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